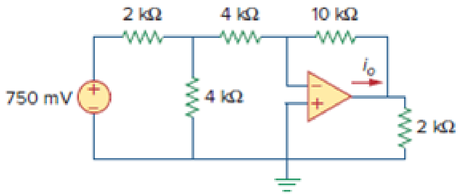
Chapter 5, Problem 19P

Problem

Determine *io* in the circuit of Fig. 5.58.

Figure. 5.58:



Step-by-step solution

Step 1 of 5

Draw the circuit diagram as shown in Figure 1.

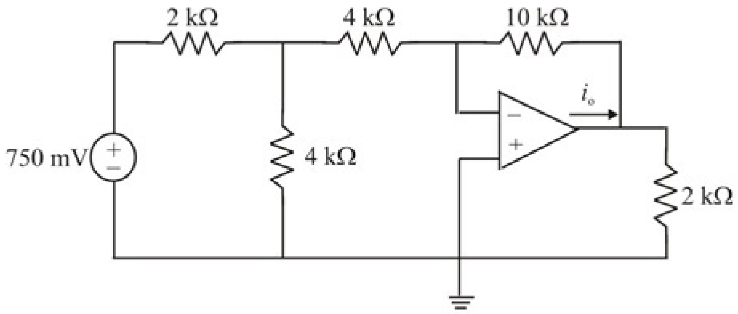


Figure 1

Step 2 of 5

The circuit is labeled as shown in Figure 2.

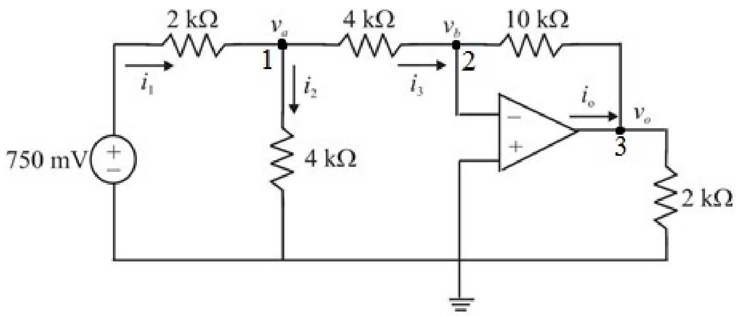


Figure 2

Step 3 of 5

Applying Kirchhoff's current law to node 1.

$$i_1 = i_2 + i_3$$
$$\left[\frac{(750\times10^{-3})-v_a}{2}\right] = \left(\frac{v_a-0}{4}\right)+\left(\frac{v_a-v_b}{4}\right)$$
$$1.5-2v_a = v_a + v_a - v_b$$
$$4v_a - v_b = 1.5$$

Therefore,

$4v_a - v_b = 1.5$ …… (1)

But, $v_b = 0$ since, non-inverting terminal is grounded.

Substitute 0 for v_b in equation (1).

$$4v_a - 0 = 1.5$$
$$v_a = 0.375\,\text{A}$$

Therefore, $v_a = 0.375\,\text{V}$

Apply Kirchhoff's current law at node 2.

$$i_3 = i_o$$
$$\frac{v_a - v_b}{4} = \frac{v_b - v_o}{10}$$
$$5v_a - 5v_b = 2v_b - 2v_o$$
$$2v_o = 7v_b - 5v_a$$

Therefore,

$2v_o = 7v_b - 5v_a$ …… (2)

Step 4 of 5

But, $v_b = 0$ since non-inverting terminal is grounded.

Substitute 0 for v_b in equation (2).

$$\begin{aligned} 2v_o &= 7v_b - 5v_a \\ &= 7(0) - 5(0.375) \\ v_o &= -0.9375\,\text{V} \end{aligned}$$

Step 5 of 5

Determine current i_o.

Apply Nodal analysis at node-3.

$$i_o = \frac{v_o - v_b}{10\times10^3} + \frac{v_o - 0}{2\times10^3}$$

But, $v_b = 0$ since non-inverting terminal is grounded.

Substitute 0 for v_b.

$$i_o = \frac{v_o}{10\times10^3} + \frac{v_o}{2\times10^3}$$

Substitute ◊◊◊◊◊◊◊ for v_o.

$$\begin{aligned} i_o &= \frac{-0.9375}{10\times10^3} + \frac{-0.9375}{2\times10^3} \\ &= -0.5625\times10^{-3} \\ &= -562.5\ \mu\text{A} \end{aligned}$$

Therefore the current i_o is $\boxed{-562.5\,\mu\text{A}}$.